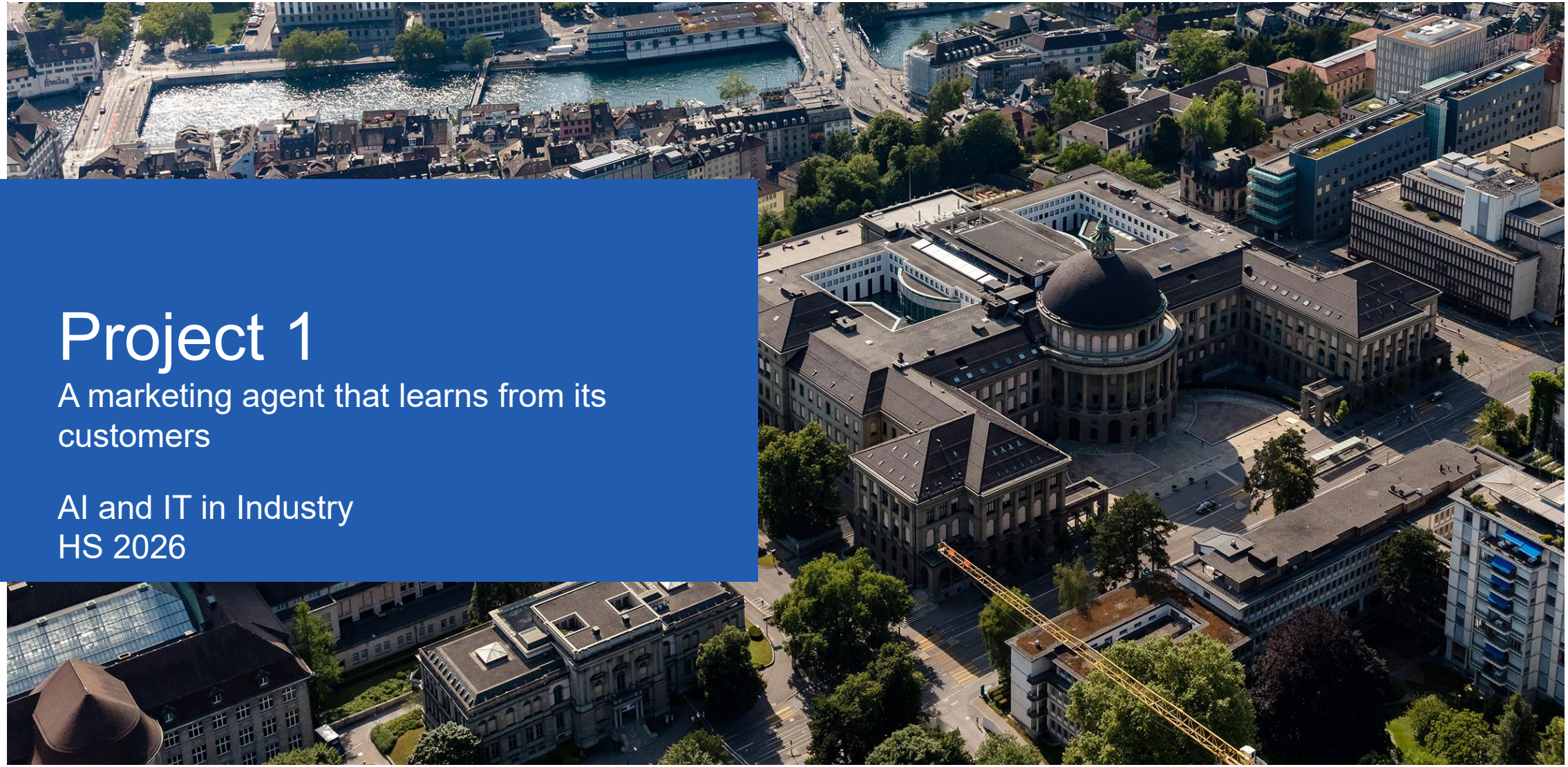
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Project 1

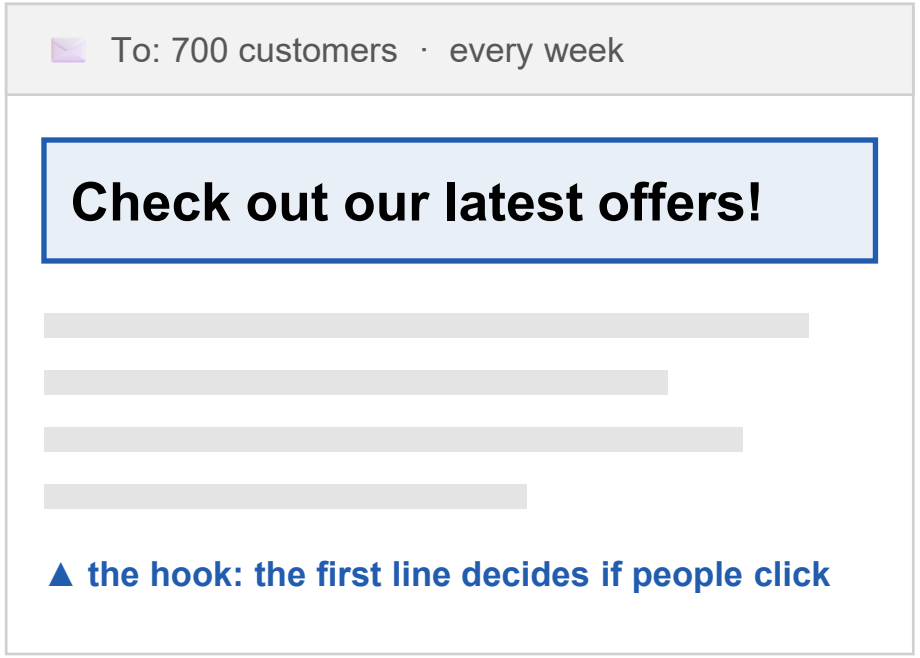
A marketing agent that learns from its customers

AI and IT in Industry
HS 2026

ETH zürich

Motivation

The project: an AI marketing agent



- You work in the marketing team of an **online clothing shop** in Switzerland with **700 customers**.
- Every week the shop sends an email. Whether a customer visits the shop depends mostly on its first line: the **hook**.
- You build an **AI marketing agent** that writes hooks and improves them with **OPRO** (Optimization by PROmpting).
- You never write a formula: you steer the agent with **instructions in plain English**.

Why is this hard?



Different customers, different wishes

A new customer, a big spender and an unhappy customer do not react to the same email.



You cannot read their minds

You only see how many customers visited the shop, plus a little customer feedback.



No formula for a good email

"Good" is a matter of language and taste, not something you can compute.

Idea: let a language model try hooks, show it how each one did, and ask it for a better one. Repeat. That is **OPRO**.

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The World

What is in the World?



Customers

700 customers of the shop, with age, number of orders, order value, last order and satisfaction.



Segments

Five groups that want different things.

N H R L A



Hidden customers

For each segment, a language model plays a customer with a secret description.

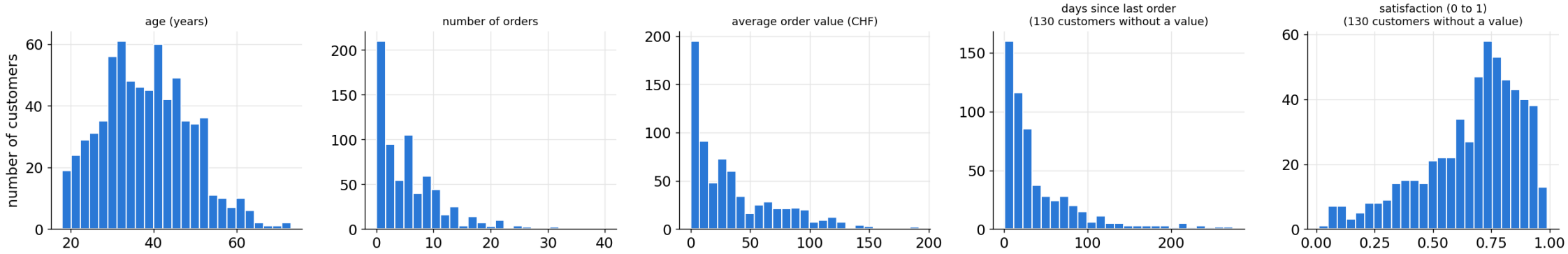


Marketing agent

Writes one hook per segment and learns from the results. **You build this.**

Everything is **simulated**, so you can run a whole marketing season in a few minutes.

Meet your customers



Every customer is one **dictionary** in Python:

```
{'name': 'Marco Roth', 'age': 39,
 'n_purchases': 4, 'avg_order_value': 28.7,
 'days_since_last_purchase': 53,
 'satisfaction': 0.71}
```

- Most customers order rarely and spend little; a few spend a lot.
- **130 customers have never ordered**: no last order, no satisfaction.
- Some have not ordered for months, or were unhappy last time.

Five customer segments

N New

Signed up recently but have not bought anything yet.

130 customers

H High-value

Buy often and spend a lot per order.

130 customers

R Regular

Buy from us now and then.

150 customers

L Low-budget

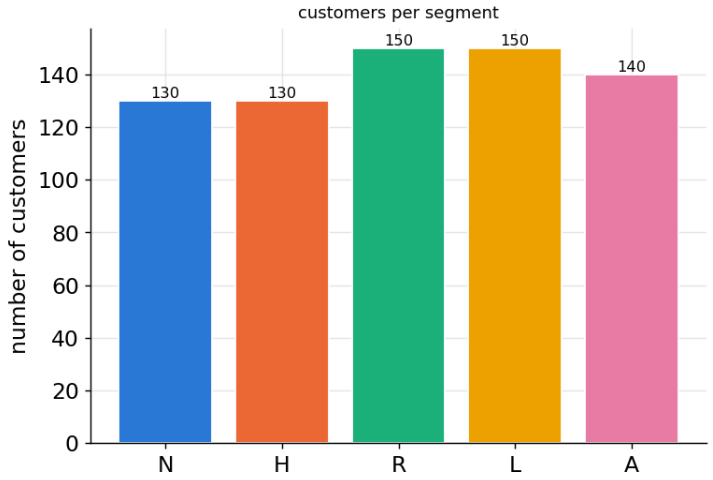
Buy cheap items and spend little per order.

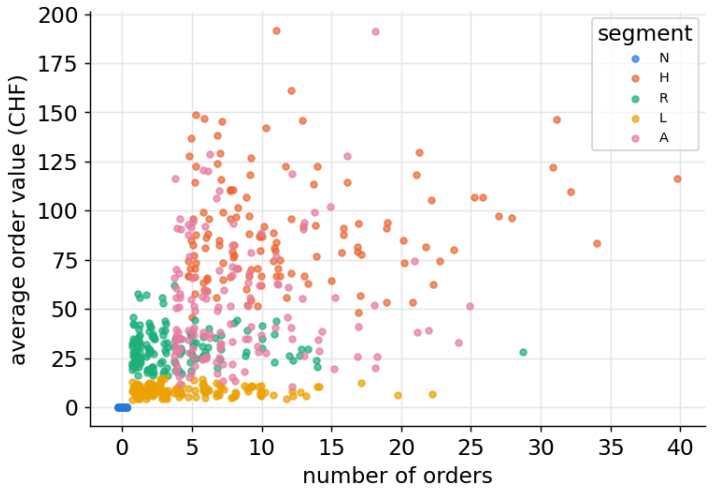
150 customers

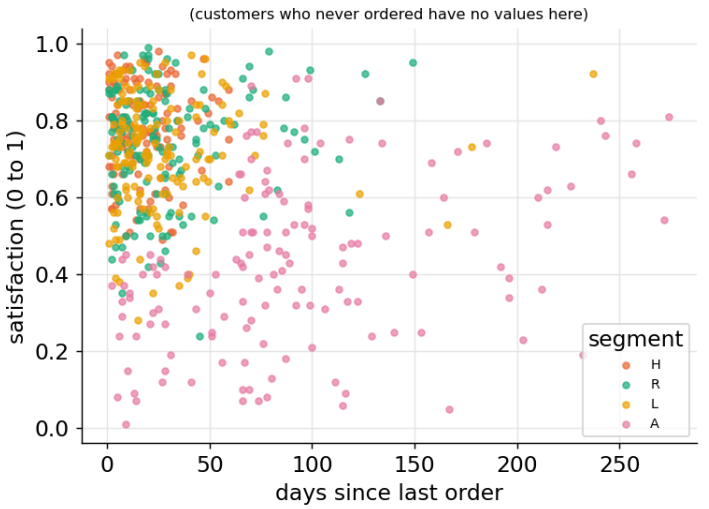
A At-risk

Used to buy regularly; gone quiet or unhappy.

140 customers







The twist: a hidden customer for every segment



The shop's current hook, scored by each hidden customer:

N 0.19 **H 0.23** **R 0.28** **L 0.20** **A 0.27**

For you it is a **black box**: text goes in, a number comes out. The score says how strongly the customer leans towards YES.

What you see, and what stays hidden

👀 You see

- The **customer data**: 700 rows of numbers.
- A short **description** of each segment, e.g. *"Buy cheap items and spend little per order."*
- **Customer feedback** from surveys and support tickets, e.g. *"Shipping costs make small orders not worth it."*
- The **visit rate** of every campaign you send.

🔒 Hidden

- What each segment's customer **really wants**, and what puts them off.
- Which hook would be the best one.

How does the World work?

1

The shop sends a campaign

Every customer gets the hook of their segment.

✉️ **Hook per segment**

→

2

The hidden customer reads it

The language model gives the hook a score between 0 and 1.

🎭 **Score**

→

3

Each customer flips a biased coin

A better score means a better chance to visit, but no guarantee.

🎲 **Visit probability**

→

4

Count the visits

The share of each segment that visited: the number the agent tries to push up.

📊 **Visit rate**

visit probability = 5% + 60% × score

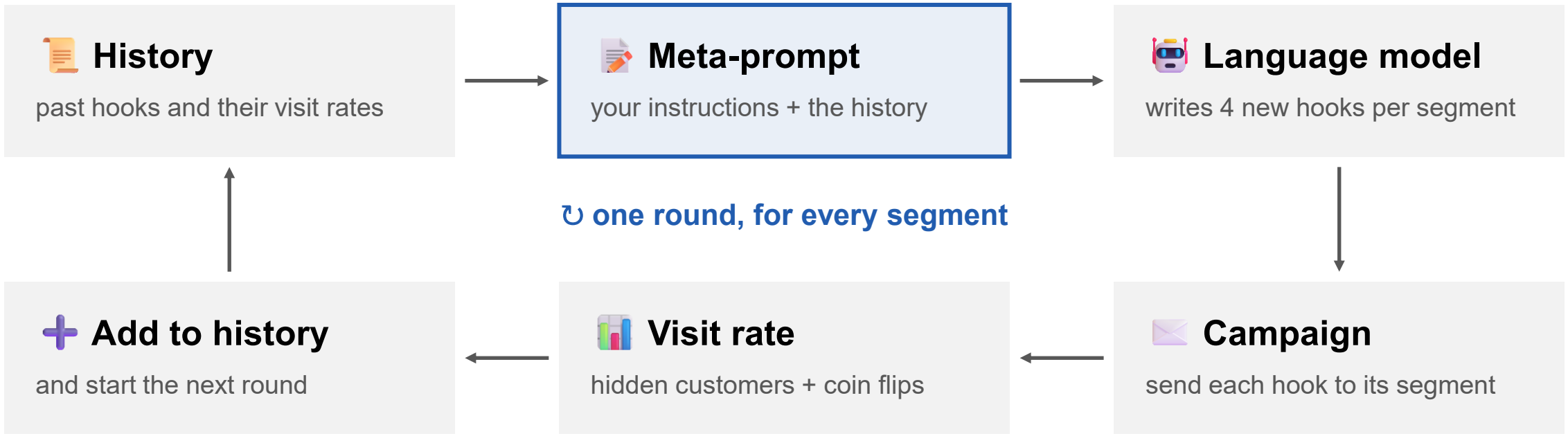
A hook with score 0.5 gives each customer a 5% + 60% × 0.5 = **35%** chance to visit. Even a perfect hook (score 1) brings at most **65%**.

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The agent: OPRO

Optimization by PROmpting

OPRO: Optimization by PROmpting

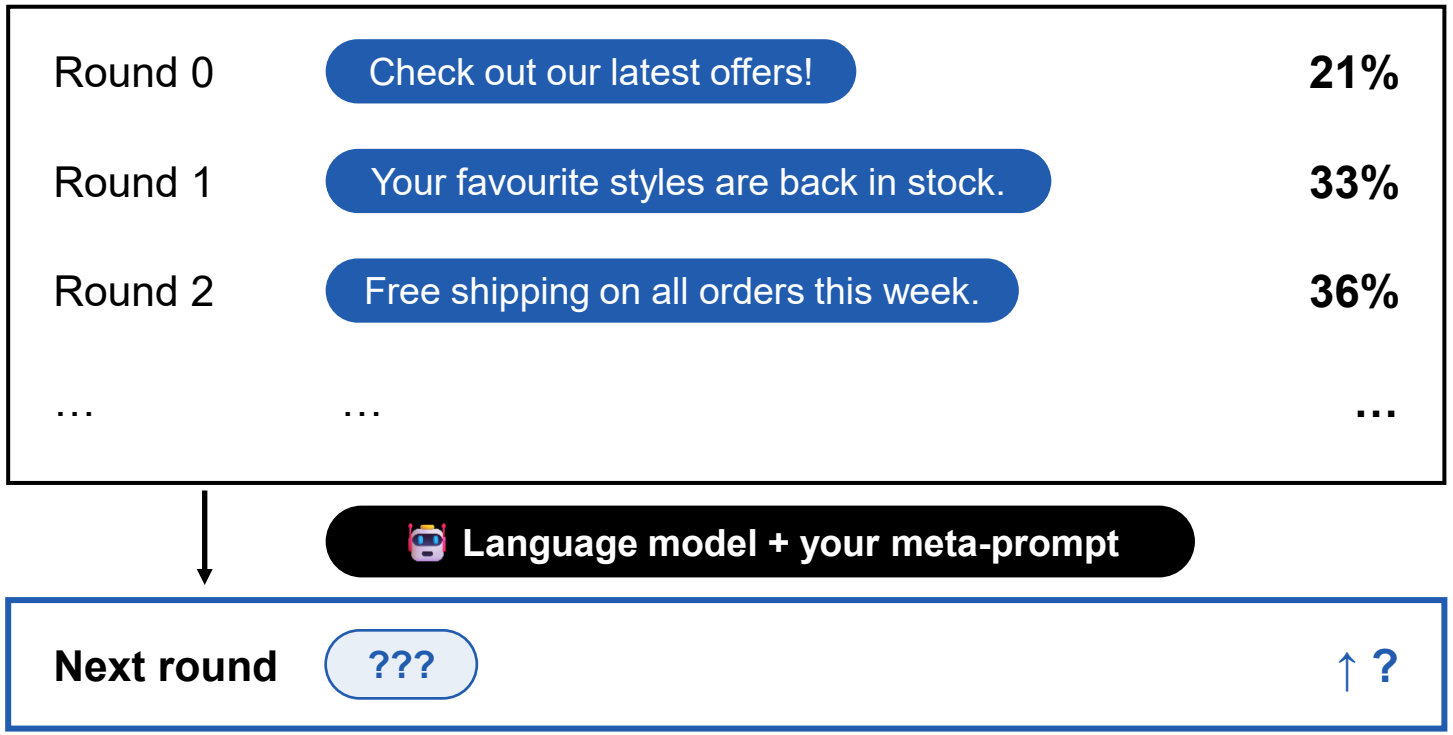


Goal: step by step, discover the hooks that make the most customers of each segment visit the shop.

Yang et al., *Large Language Models as Optimizers*, Google DeepMind, ICLR 2024 (arXiv:2309.03409)

OPRO, round by round

History of one segment, sorted from worst to best (the best one last)



- Each round the model reads the **k best** past hooks with their visit rates.
- It writes **4 new hooks** per segment; all of them are sent and measured.
- **6 rounds**, all five segments at once.

The meta-prompt: your main lever

🎯 TODO 8

You start from this (it works, but barely):

```
Write a marketing email hook.

Past hooks:
{past_hooks}

Reply with the hook only.
```

There is no single right answer. Different meta-prompts give different results, and you explain your choices in the defence.

Ideas to make it better:

- Give the model a **role** and explain the situation.
- Tell it **who the customers are** (the segment description).
- Pass on the **customer feedback**: the only real clue.
- Show the **past hooks** and say what the numbers mean.
- Ask it to **think first**: what do these customers care about, what stops them?
- Ask for something **new and concrete**, not a small variation.
- Set **rules**: length, no placeholders like [Name].

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Workflow

What you do in the notebook

10 TODOs in 5 parts



PART 1
Meet your customers
TODO 1
Write hooks by hand and see how customers react.

PART 2
Segments
TODO 2–3
Sort customers with five rules and count them.

PART 3
Campaign
TODO 4–5
Visit probability and visit rate per segment.

PART 4
OPRO agent
TODO 6–8
Show the history, clean the answer, write the meta-prompt.

PART 5
Run & experiment
TODO 9–10
Compare with the baseline, then change one thing.

- Short **Python functions** over lists and dictionaries.
- After most TODOs, a **check cell** tells you if it works.

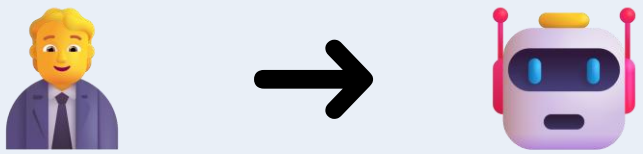
```
✅ score 0 gives just the base rate: expected 0.05, got 0.05
✅ score 1 gives base rate + lift: expected 0.65, got 0.65

All checks passed. 🎉
```

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Summary

Project 1: LLM-based marketing agent



From writing every hook yourself
to steering an agent that writes them

Goals

- Create an **LLM agent** that **proposes hooks** for each customer segment
- Track the **performance of each hook** on simulated customers
- Find the **best hooks** with **Optimization by PROmpting**

Skills

- **LLM usage and prompting**
- Setting up an **optimization problem** with a goal written in words
- Reading **noisy results** like a marketing analyst

Result

- **An LLM-powered marketing agent!**